Mapping Persistent Urban Surface Heat and Environmental Vulnerability in New York City

A neighborhood-scale geospatial analysis | Summers 2022–2025

RESEARCH REPORT • Google Earth Engine • Python • GIS

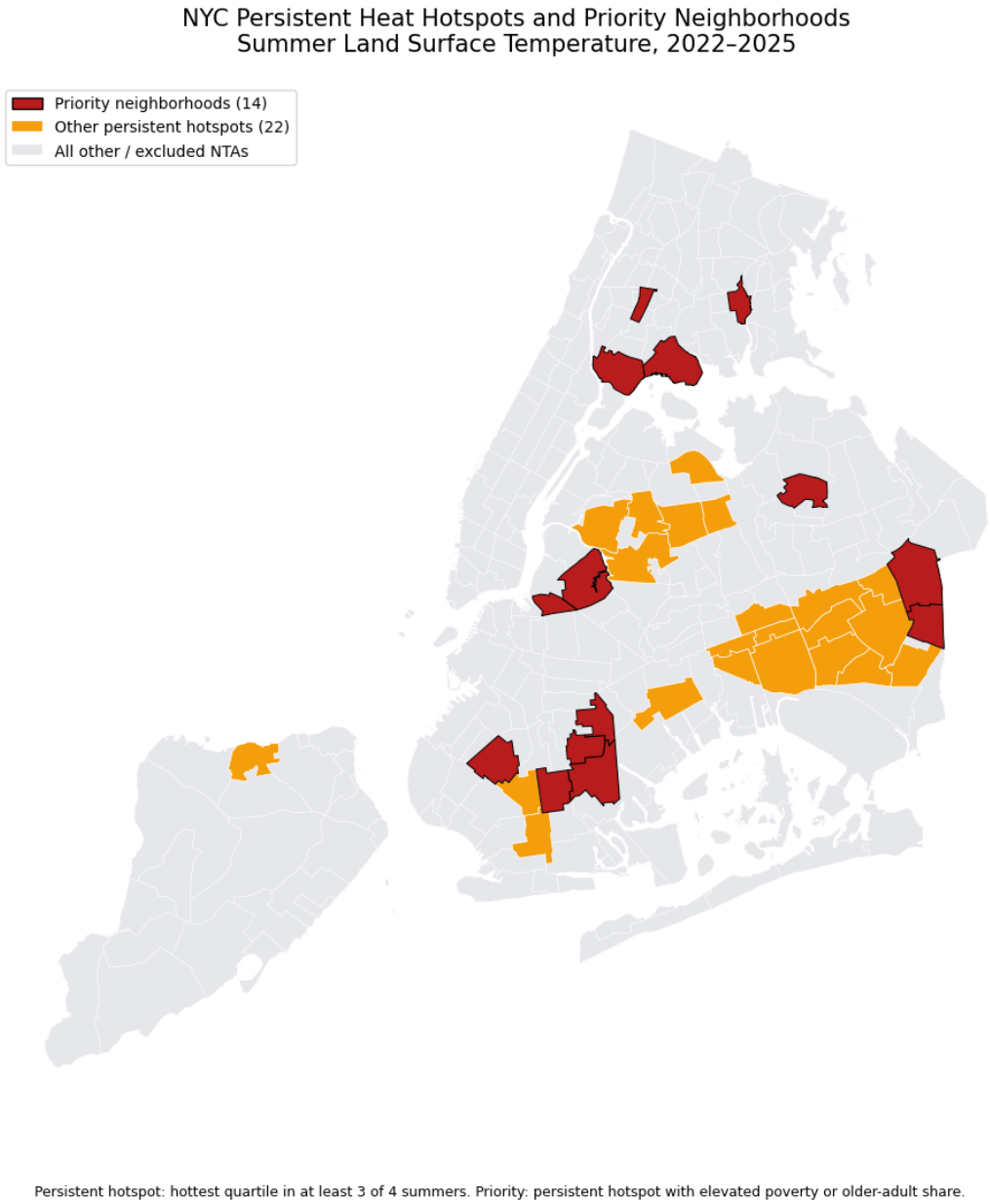


Figure 1. Persistent heat hotspots (orange) and selected social-vulnerability priority neighborhoods (dark red). Gray indicates other or excluded NTAs.

Executive summary

This project integrates Landsat-derived summer land surface temperature (LST), vegetation greenness, impervious cover, and selected socioeconomic indicators to screen New York City neighborhoods for persistent surface heat and overlapping vulnerability. The analytical sample includes 197 standard Neighborhood Tabulation Areas (NTAs), drawn from 262 total NTAs.

197	36	14
Standard NTAs analyzed	Persistent heat hotspots	Priority neighborhoods

Persistent hotspots were defined as NTAs in the hottest annual quartile in at least three of four summers. Fourteen of these also met a high-poverty or high-older-adult-share criterion. An additional permanent-water mask changed average NTA temperatures by about 0.03°C per year, but changed no persistent-hotspot classifications.

Research questions

Where does summer surface heat persist? How do vegetation and imperviousness relate to LST? Where do persistent hotspots overlap with selected indicators of social vulnerability? How stable are results to an alternative water mask?

Study design and data

Component	Approach / source
Geography	2020 NYC NTA boundaries; 197 standard NTAs
Surface temperature	Landsat 8/9 Collection 2 Level-2; June–August 2022–2025; Google Earth Engine
Vegetation	Landsat-derived NDVI, averaged over 2022–2025
Built environment	NLCD 2021 impervious surface percentage
Social indicators	ACS 2024 five-year poverty and age 65+; tract-to-NTA crosswalk
Sensitivity	JRC Global Surface Water baseline plus ESA WorldCover permanent-water exclusion

Methodology

Quality-filtered Landsat surface-temperature observations were scaled to degrees Celsius and aggregated into month-balanced summer composites. Neighborhood means were calculated from spatial reductions. Each year, the hottest quartile of standard NTAs was flagged; persistence required at least three flagged summers. Elevated social vulnerability was screened using the upper quartile of poverty (approximately 23.42%) or population aged 65+ (approximately 19.82%). This rule-based screen avoids assigning arbitrary weights to unlike indicators.

Geographic distribution of surface heat

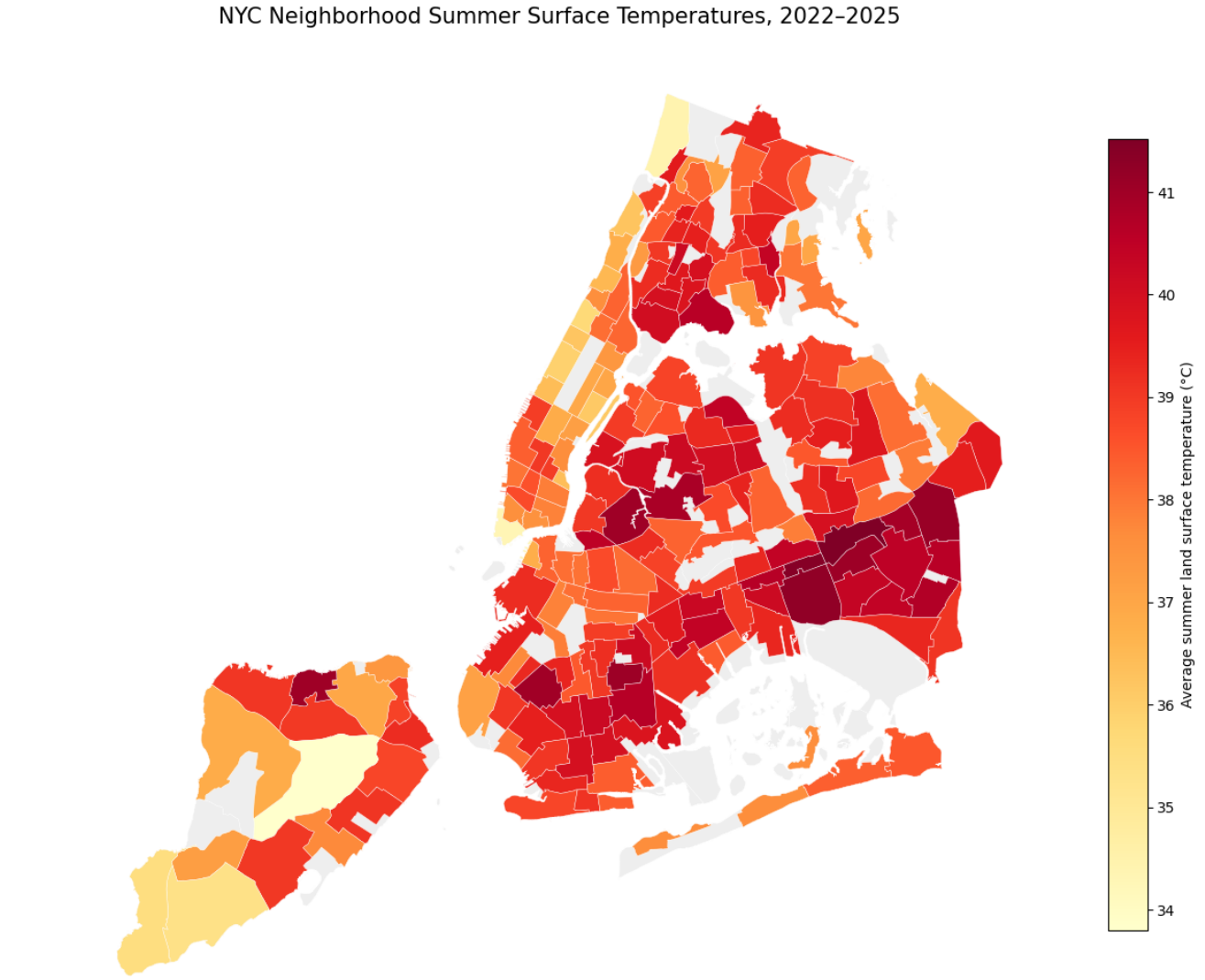


Figure 2. Four-year average summer land surface temperature across standard NTAs. Gray areas are outside the 197-NTA analytical sample; LST is not ambient air temperature.

Persistent hotspot patterns

Measure	Persistent hotspots (n=36)	Other NTAs (n=161)
Mean LST	40.543°C	38.375°C
Mean impervious cover	82.874%	77.725%
Mean NDVI	0.262	0.297
Mean poverty	17.986%	17.361%
Mean population aged 65+	16.130%	16.978%

The LST difference is expected because hotspot status is temperature-defined. Differences in impervious cover and NDVI are descriptive patterns that warrant further investigation, not causal estimates.

Land cover and temperature relationships

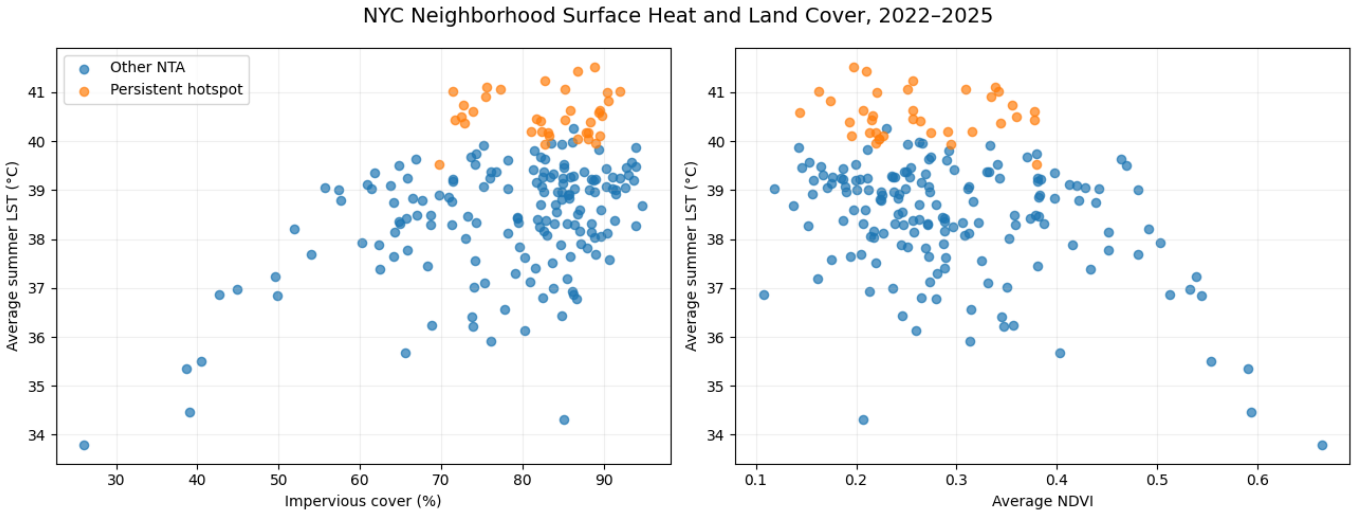


Figure 3. Neighborhood-average summer LST versus impervious cover and NDVI; orange points indicate persistent hotspots.

Indicator	Spearman correlation with mean LST
Impervious cover	+0.283
NDVI	−0.238
Poverty rate	+0.182
Population aged 65+	−0.171

All four associations are modest. In particular, a negative neighborhood-level correlation for older-adult share does not imply lower heat-health risk among older residents. Surface heat, demographic susceptibility, and actual exposure are distinct constructs.

Neighborhood priority profiles

Hunts Point: 40.60°C LST, 89.35% impervious cover, NDVI 0.144, and 28.59% poverty; a candidate for detailed land-use, shade, and industrial infrastructure assessment.

Borough Park: 41.00°C LST, 90.48% impervious cover, and 33.70% poverty; a candidate for street-scale shade and greening feasibility assessment.

Cambria Heights: 40.62°C LST and 24.04% aged 65+, despite NDVI 0.378; illustrates why greenness alone cannot determine social vulnerability.

Fourteen priority neighborhoods

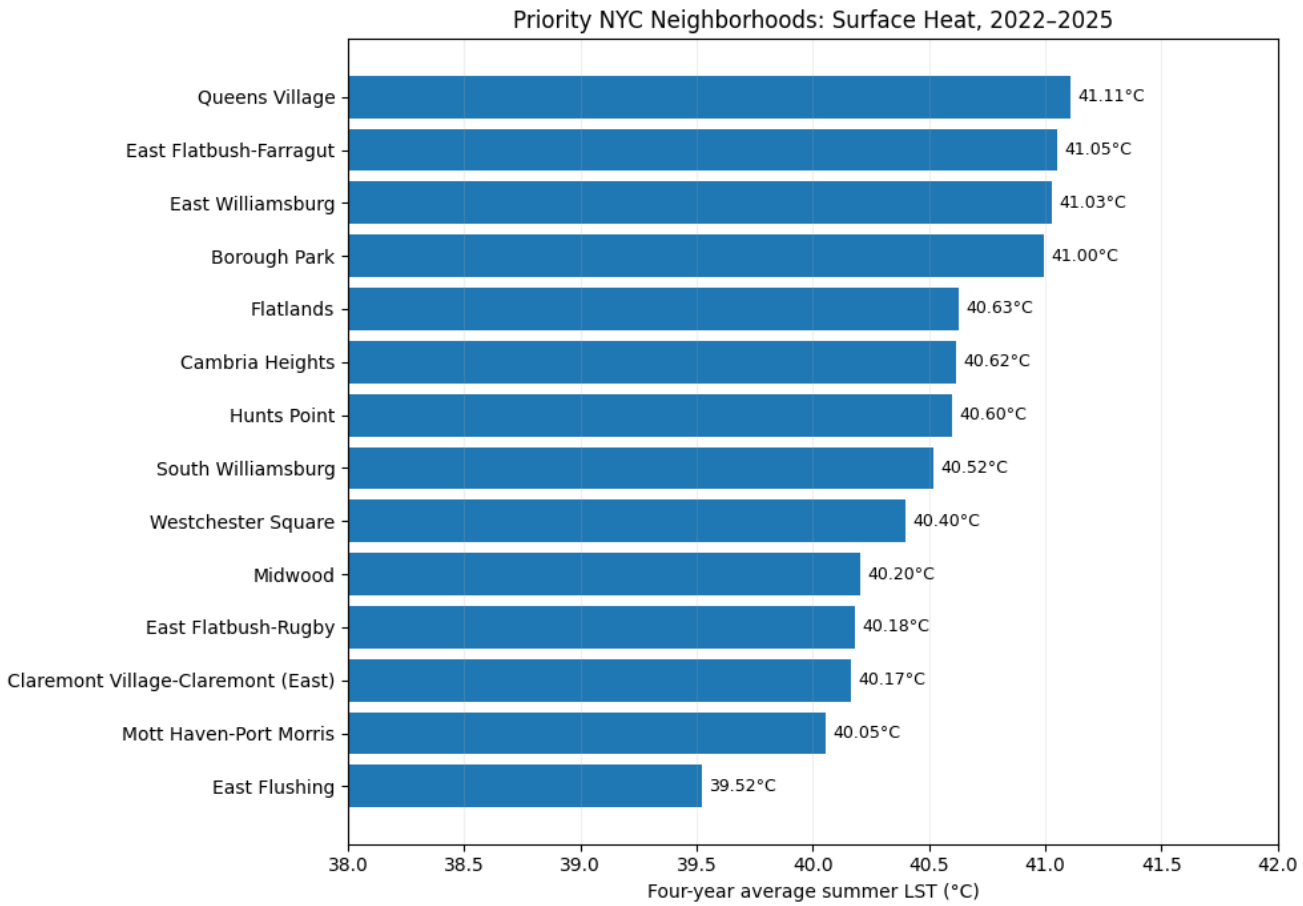


Figure 4. Four-year average LST of selected priority NTAs. The chart axis begins at 38°C; interpret differences using the numeric labels.

Priority means a persistent heat hotspot meeting the upper-quartile poverty or older-adult-share screening rule. It is not a causal estimate, a full vulnerability index, or a ranking of intervention cost-effectiveness.

Masking sensitivity and validation

Summer	Mean absolute change	Maximum absolute change
2022	0.0300°C	0.3045°C
2023	0.0295°C	0.3950°C
2024	0.0289°C	0.2776°C
2025	0.0306°C	0.3748°C

All 262 NTAs were compared in each year. The reconstructed baseline matched the original annual NTA means exactly at the reported precision. After adding the permanent-water exclusion, all 36 persistent hotspots retained their status, and all 14 selected priorities retained their number of hottest-quartile years.

Planning implications

1. Target field assessment, not immediate prescriptions. Use the screen to select places for street-level shade audits, canopy inventories, pavement surveys, and public-space assessments.

2. Distinguish infrastructure contexts. Industrial districts, dense residential streets, and lower-density neighborhoods may require different design options and feasibility checks.

3. Layer additional exposure and equity evidence. Evaluate pedestrian activity, cooling access, indoor conditions, health burden, and community input before deciding on investments.

Limitations

Landsat LST measures surfaces at satellite overpass, not air temperature, indoor conditions, or personal heat exposure. Clouds, observation timing, mixed land cover, and neighborhood averaging introduce uncertainty. Imperviousness uses 2021 data, while ACS 2024 five-year estimates and 2022–2025 satellite data have different temporal frames. The socioeconomic screen uses only poverty and older-adult share. Ecological associations are not causal. The water-mask sensitivity check tests one specific processing decision, not all methodological alternatives.

Reproducibility and source documentation

Primary analytical environment: Python, pandas, GeoPandas, Matplotlib, geemap, and Google Earth Engine. Data products documented in the notebook include Landsat Collection 2 Level-2, NYC 2020 NTA boundaries, NLCD 2021 imperviousness, ACS 2024 five-year estimates, JRC Global Surface Water, and ESA WorldCover. For publication, add direct dataset URLs, access dates, the tract-to-NTA allocation rule, and a frozen notebook/software environment.

Conclusion

The analysis identified 36 persistent NYC surface heat hotspots and 14 locations where persistent heat overlaps with selected socioeconomic vulnerability indicators. Higher imperviousness and lower vegetation generally accompanied higher neighborhood LST, though associations were modest. Classifications were unchanged by the tested additional water mask. The project provides a reproducible screening basis for further urban heat-resilience and infrastructure analysis.

Source note

Findings and embedded figures were drawn from the uploaded Colab notebook and the validated results supplied during the analysis. This report has not independently rerun Earth Engine computations or audited the original external datasets.